

B Prompt Templates and Examples

B.1 Prompt Template of Note Construction

The prompt template in Note Construction: P_{s1}

Generate a structured analysis of the following content by:

1. Identifying the most salient keywords (focus on nouns, verbs, and key concepts)
2. Extracting core themes and contextual elements
3. Creating relevant categorical tags

Format the response as a JSON object:

```
{
  "keywords": [ // several specific, distinct keywords that capture
    key concepts and terminology // Order from most to least important //
    Don't include keywords that are the name of the speaker or time // At
    least three keywords, but don't be too redundant. ],
  "context": // one sentence summarizing: // - Main topic/domain // -
    Key arguments/points // - Intended audience/purpose ,
  "tags": [ // several broad categories/themes for classification //
    Include domain, format, and type tags // At least three tags, but
    don't be too redundant. ]
}
```

Content for analysis:

B.2 Prompt Template of Link Generation

The prompt template in Link Generation: P_{s2}

You are an AI memory evolution agent responsible for managing and evolving a knowledge base.

Analyze the the new memory note according to keywords and context, also with their several nearest neighbors memory.

The new memory context:

{context} content: {content}

keywords: {keywords}

The nearest neighbors memories: {nearest_neighbors_memories}

Based on this information, determine:

Should this memory be evolved? Consider its relationships with other memories.

B.3 Prompt Template of Memory Evolution

The prompt template in Memory Evolution: P_{s3}

You are an AI memory evolution agent responsible for managing and evolving a knowledge base.

Analyze the the new memory note according to keywords and context, also with their several nearest neighbors memory.

Make decisions about its evolution.

The new memory context:{context}

content: {content}

keywords: {keywords}

The nearest neighbors memories:{nearest_neighbors_memories}

Based on this information, determine:

1. What specific actions should be taken (strengthen, update_neighbor)?

1.1 If choose to strengthen the connection, which memory should it be connected to? Can you give the updated tags of this memory?

1.2 If choose to update neighbor, you can update the context and tags of these memories based on the understanding of these memories.

Tags should be determined by the content of these characteristic of these memories, which can be used to retrieve them later and categorize them.

All the above information should be returned in a list format according to the sequence: [[new_memory],[neighbor_memory_1],...[neighbor_memory_n]]

These actions can be combined.

Return your decision in JSON format with the following structure: {{

"should_evolve": true/false,

"actions": ["strengthen", "merge", "prune"],

"suggested_connections": ["neighbor_memory_ids"],

"tags_to_update": ["tag_1",... "tag_n"],

"new_context_neighborhood": ["new context",...,"new context"],

"new_tags_neighborhood": [{"tag_1",...,"tag_n"},...["tag_1",...,"tag_n"]],
}}

B.4 Examples of Q/A with A-MEM

Example:

Question 686: Which hobby did Dave pick up in October 2023?

Prediction: photography

Reference: photography

talk start time:10:54 am on 17 November, 2023

memory content: Speaker Davesays : Hey Calvin, long time no talk! A lot has happened. I've taken up photography and it's been great - been taking pics of the scenery around here which is really cool.
memory context: The main topic is the speaker's new hobby of photography, highlighting their enjoyment of capturing local scenery, aimed at engaging a friend in conversation about personal experiences.

memory keywords: ['photography', 'scenery', 'conversation', 'experience', 'hobby']

memory tags: ['hobby', 'photography', 'personal development', 'conversation', 'leisure']

talk start time:6:38 pm on 21 July, 2023

memory content: Speaker Calvinsays : Thanks, Dave! It feels great having my own space to work in. I've been experimenting with different genres lately, pushing myself out of my comfort zone. Adding electronic elements to my songs gives them a fresh vibe. It's been an exciting process of self-discovery and growth!

memory context: The speaker discusses their creative process in music, highlighting experimentation with genres and the incorporation of electronic elements for personal growth and artistic evolution.

memory keywords: ['space', 'experimentation', 'genres', 'electronic', 'self-discovery', 'growth']

memory tags: ['music', 'creativity', 'self-improvement', 'artistic expression']